

Subashree Venkatesan Sundharesan

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EDUCATION

Master of Science in Computer Science

Indiana University - Bloomington, USA

CGPA: 3.7/4

August 2024 – May 2026

Coursework: Applied Algorithms, Advanced Database Concepts, Applied Machine Learning, Software Engineering, Native Mobile Application Development (Android and iOS), Applied Database Technologies

Bachelor of Engineering in Computer Science

Anna University - Chennai, India

CGPA: 8.5/10

August 2018 – May 2022

Coursework: Object Oriented Programming, Data Structures, DBMS, Mobile Computing, Cloud Computing, Operating Systems

WORK EXPERIENCE

Software Engineer

Amadeus

August 2022 – August 2024

Bangalore, India

- Developed reusable UI components for Air, Hotel, and Rail booking flows and redesigned a comprehensive user notification module ensuring seamless compatibility across 10,000+ production systems. Used Java, Angular, XSLT and Closure.
- Implemented sustainability-focused features, such as CO2 emissions for flights & recommending eco-friendly hotels, for environmental awareness. Optimized Oracle SQL stored procedures, resulting in a 42% increase in page load efficiency.
- Led repository migration from BitBucket to GitHub, modernizing version control systems and enhancing DevOps workflows.
- Developed Spring Boot microservices for re-encryption processes, ensuring data security and improved performance.
- Achieved over 90% JUnit test coverage, ensuring high code quality and reliability, and deployed applications to the OpenShift cloud platform via Jenkins for automated testing and seamless production rollouts.
- Automated the developer environment setup process using scripts, reducing onboarding time by 75% (10 to 2.5 days)
- Mentored 6+ new hires and interns by providing technical guidance and helped with setup, KTs to ensure team success.

Software Engineer Intern

Amadeus

March 2022 – July 2022

Bangalore, India

- Modernized a 20-year-old translation tool, addressing key pain points and improving efficiency with 30% performance boost.
- Redesigning front-end from servlet UI into a single page web application using Angular and Angular Material, incorporated NgRx(Redux) for better state management
- Rebuilt back-end logic from Java Servlets to Spring Boot and created REST APIs to perform CRUD operations
- Created unit tests using JUnit5, Jasmine & Karma and improved overall performance by 30%

Research Assistant

Sri Sivasubramaniya Nadar College of Engineering

July 2021 – June 2022

Chennai, India

- Worked under the guidance of Professor Dr. Felix Enigo V S and proposed a tool that can be used for visualization and investigation of real-world crime networks, based on Social Network Analysis
- Created dynamic network graphs using Python, JavaScript & frameworks such as NetworkX, Pyvis, Vis.js
- Computed centrality measures, detected communities, added searching and filtering functionality based on names, crimes, locations of the criminals and visually represented them in the network
- Used path finding and shortest path algorithms to find the links and people between any two criminals
- Developed a web application that could be used to gain insights from the analysis made for any given dataset. Dataset consists of data that is cleaned and extracted from word documents used by the Police Department
- Demonstrated the tool at the State Crime Records Bureau, Tamilnadu, India.

Frontend Developer Intern

Center for Development for Advanced Computing (C-DAC)

April 2021 – June 2021

Chennai, India

- Built a tool for the Tamilnadu (State in India) Police Department to help them analyze crimes across the state
- Created the dashboard such that it generates dynamic graphs and reports based on the conditions given by the user, across different timelines and criminal cases using JavaScript, Chart.js, jQuery, HTML, CSS, Bootstrap.
- Constructed different queries for the conditions and made API calls to fetch the required data from backend
- Coordinated with the Big Data Team and improved the speed and functionality of the dashboard

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, SQL, JavaScript, TypeScript

Frontend: Angular, React, Tailwind, NgRx(Redux), HTML, CSS, SCSS Angular Material, Vue.js Bootstarp, Kotlin, Flutter

Backend: Spring Boot, JUnit, MySQL, Oracle SQL, MongoDB, PostgreSQL, Node.js, SQLite, NOSQL, Relational Database

Machine Learning: Numpy, Pandas, Matplotlib, Scikitlearn, TensorFlow, NetworkX, Jupyter Notebook, Google Colab

Developer Tools: Git, GitHub, BitBucket, Jira, Confluence, Postman, Docker, Podman, RabbitMQ, VS Code, IntelliJ, Android Studio, Eclipse, Visual Studio, Microsoft Azure, Jenkins, RedHat OpenShift, SonarQube, Kibana, Shell Scripting, Unix, Linux, PowerShell, JSON, XML, SOAP, CI/CD, UML

Others: Word, Excel, PowerPoint, Outlook, Teams, Agile Methodologies, SAFe Practices, Scrum, SDLC Debugging, Code Review, Technical Documentation

PROJECTS

Stories App  | *React.js, TailwindCSS, Node.js, Express.js, MongoDB Atlas, JWT, RESTful APIs*

Developed a full-stack MERN application similar to Instagram stories, enabling users to create, edit, manage stories with features like image uploads, JWT-based authentication, pinned highlights, and advanced search/filtering by date, location, or keywords.

AI Fitness Trainer  | *Python, Numpy, OpenCV, MediaPipe*

Counts the number of reps in a workout by detecting land marks and angles in a human body. Integrated with live camera.

Weather App  | *Flutter, Dart, Android Studio, REST APIs, Lottie Animations, iOS*

Cross-platform mobile app fetches real-time weather updates for current or specified locations via OpenWeatherMap API.

Recipe Finder  | *React, MaterialUI, JavaScript, HTML, CSS*

Responsive web application which displays recipes based on the searched ingredient or dish. Uses Edamam Food API.

Mutual Authentication Based Secure Transmission of E-Healthcare Informations  | *AVISPA, Scyther, BAN Logic*

Devised a new two-way protocol to transfer healthcare data along with proof for robustness using formal & informal analysis

CERTIFICATIONS

SQL for Data Science  | *UC Davis, Coursera*

Data Analyst with Python Track (Specialization)  | *DataCamp*

SAFe 5 Practitioner  | *Scaled Agile Inc*

Finance for Engineering Professionals  | *Emory University*

EXTRA CURRICULAR ACTIVITIES

IU Dining and Hospitality – Supervisor

- Managed labor distribution, task assignments, and staff training to ensure compliance with standards
- Handled opening/closing procedures, cash reconciliation, and operational support during peak periods
- Resolved customer concerns to maintain positive relations and mentored part-time employees, fostering skill development and promoting a collaborative work environment
- Conducted inventory counts and represented management to maintain smooth operations

Youth Red Cross – Member

- Volunteered for a two-day village camp. Took Census at rural areas to understand the people's health care requirements and taught yoga to school children
- Demonstrated the need for wearing seat belts, helmets and following traffic rules to the general public at signal junctions in Thiruvanniyur, Chennai
- Took part in beach, lake cleanup programs and awareness rallies

Entrepreneurship Development Cell – Core Committee Member

- Organized and coordinated SSN Youth Conference and SSN Mela (annual college fair)
- Been part of the sponsorship, marketing, and ticketing teams during the yearly fair and conferences

Bharat Scouts and Guides – Guide

- Part of a three-day Guide's camp: Camped and stayed near forest areas, Underwent intense physical training, Cooked food for fellow guides and staff, Took part in trekking and self-defense classes, Was awarded one among the Top 5 Guides
- Volunteered for Annual Day and Sports Day events